

Team Root Matrix: Mary Ann Mastri, Cesar Noriega, Luiz Paludo, Nichelle Graf
A04 Deep Learning for an 11-Year-Old – Team Contributions
ITAI 2376 – Deep Learning in AI
Professor Viswanatha Rao

A04 - Deep Learning for an 11-Year-Old Team Contributions

Mary Ann:

I created an idea that evolved. It first started out purely academic and then developed into turning the idea into a game. The original idea was refined as Claude AI assisted in generating code and turned the idea into a game. The initial game was improved upon. After a few iterations of developing the game, which was just a spur-of-the-moment idea, a final version was created. There was input and collaboration from the team to flesh out the academic basis of the game.

Cesar:

This project was helping my group to understand and explain entropy and cross-entropy in a simple way. I participated in a discussion about how to use the Dragon Egg Cave game to make these concepts easier for an 11-year-old to understand. I also helped review the explanations and make sure they were connected to Deep Learning and classification. I shared ideas with my teammates and reviewed the content to make sure the information was clear and easy to follow. Working on this part of the project helped me better understand how uncertainty, probability, prediction, and loss are connected in Deep Learning.

Luiz:

My main contribution was on the research and testing side of the project. After the game was finalized, I looked into how entropy and cross-entropy are actually used in classification models, so that the way we presented them in the Dragon Egg Cave stayed accurate even after we simplified the language for an 11-year-old. Once the game was built, I went through each activity myself — the confusion meter, the warmth sliders, and the Human vs. Dragon-Bot comparison — to check that what was happening on screen actually matched what we were claiming it taught. When something didn't line up cleanly with the underlying concept, I flagged it so we could adjust the wording or the explanation rather than the mechanic itself. This project taught me that simplifying a Deep Learning concept is a balancing act: you can lose accuracy pretty easily if you're not careful, and part of my role was making sure that didn't happen while the game stayed approachable for a young audience.

Nichelle:

My contributions to this project focused on evaluating the clarity and accuracy of the Dragon Egg Cave presentation, developing the written project documentation, and organizing the final GitHub repository.

I reviewed the interactive game from the perspective of the assignment requirement that the concepts be understandable to an 11-year-old. I identified areas where the wording could be simplified or clarified, particularly in the explanation of entropy, cross-entropy, prediction confidence, and the difference between a human guess and Dragon-Bot's probability-based prediction. I also reviewed the game logic and technical documentation to better understand how the entropy and cross-entropy calculations were being represented.

I prepared and revised the main project report explaining the group's topic choice, the purpose of the Dragon Egg Cave game, how the activities demonstrate entropy and cross-entropy, the design choices made for a younger audience, and the connection to deep learning classification and loss functions.

I also created and organized the GitHub repository for the final project submission. This included uploading the project report and technical documentation, developing and revising the README, adding the live game link, and organizing the repository so the project materials would be easy to review.

In addition, I completed my individual reflection and helped review the final project materials for clarity, consistency, and alignment with the assignment requirements.